

Pg - 336

Rectilinear motion

eg - 14.3 ✓ next (check) ✓
14.8

$$\begin{cases} a \cdot t^2 - 2t + s \\ \text{differentiate} \end{cases}$$

only 1 see

eg. of variable acceleration

Kinetics

14.23

✓ pulley eg.

14.24

✗

14.25

✓

14.27

1 ahead → leave ✗

3 block, 3 pulley - do ✓

Unresolved 14.23 → do

25 mm. → Theoretical

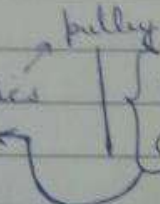
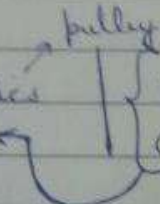
Solved age

1. Force signature
2. Factor

Variance th^m

3. Truss both methods
4. CG MT

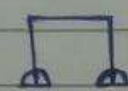
~~graphical not in syll~~

- ⑤ Kinematics ^{not waste} 
 - ⑥ Kinetics 
- Pulley.
Coff of rest → impact coll.

- ⑦ Shear force & bending moment → Cantilever or Simply supported

- ⑧ I.C., 4 bar mechanism

any method, no method specified.



~~By 25/11/2020~~
Moment - ~~xxxx~~

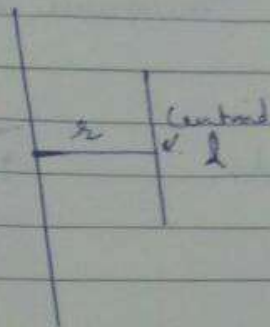
11el forces in plane ~~X~~ not in syllabus
pressure ~~X~~

CG → 2 methods ✓ both do.
↳ solid object → not in syllabus.

General forces → not in syllabus.

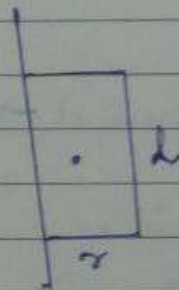
friction - do derivation, assigns. - derive
as rolling - not in syll. (iii) Attn of friction

I Pappus golden rule th^{ms}:



if rotated about axis

$$2\pi r \times l$$



dis $\times$ Area

Volume

$$\frac{2\pi r}{2} \times r \times l$$

$$= \pi r^2 l$$

II

Screw jack, belt tension $\leftrightarrow$ derivation

Friction

II

Truss

CG, moment of inertia
↓
sig

} ① ✓

Collision
full

Ch-19 - rel. motion $\times$ not compl.
20 motion of projectile $\times$

Instantaneous Centre

all solved exs. ~~ditto~~

4 bar mech^m

Ch-22

Starting 2 exs.

22.5 - pulley ex.

22.6 -

leave ahead

Ch- Shear force & bending moment
 $\rightarrow$ Photography

D.S. Kumar

Circular motion

15.3 ✓
eg- 15.4 ✗
15.5 ✓

Radial & transverse cpts - ✗ not in sys.

Kinetics

15.7a 15.9 → eg.
15.8 → ✗

leave all ahead ↘

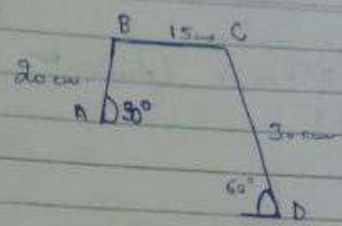
Work & Energy

Full Chapter

Pg- 449 Applⁿ of princ^{le} of conservⁿ of energy to
circular motion - ✗

eg- 17.1

✓ only
trolley, boat ✗



disk & beams } $\propto$ not in syllabus
Simple lifting machine

Tress

Analysis of tress $\propto$
 $\left\{ \begin{array}{l} P_2-208 \text{ frame related ques on tress} \\ P_2-214 \end{array} \right. \propto$ not in syll
Cable $\propto$  $\propto$ not in syllabus
 member
 cross

MI

Product of Inertia $P_2-293 \rightarrow$ not in syllabus
 displacement, rotation of Area $\rightarrow$
 Principal of virtual work $\rightarrow$

based on T_1, T_2
 eg- 7.4, 7.5 do
 (length of belt) $\propto$

reversible & irreversible machine
 (1 ques)

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Four bar + Single slider mechanism

Types of motion

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fundamental
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free-body
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Alternative
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K. TAYA

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Transmiss principle of transmission
 Polygon
 Area of base
 Radius of gyration
 Sig. of moment of Inertia
 Pt. of contraflexion
 Types of beam
 Instantaneous center of Rotation
 Hel & Lr axis - th^m

→ P-props guldiner Th^m.

$$\textcircled{1} \quad \frac{I_1}{I_2} = l^{VP}$$

$$\textcircled{2} \quad \eta = \tan(\alpha + \phi) \quad \frac{\tan \alpha}{\tan(\alpha + \phi)}$$

Angle of repose, friction
 Coefficient of Restitution